

Gravity & Jumping Mechanics

Module 2 — Session 7

Building Responsive Jumping with Physics

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Today's Mission

Today, you will build a gravity and jumping system.
You will learn how to:

- 1 Apply gravity to vertical movement
- 2 Use velocity to create a jump
- 3 Detect when the player reaches the ground
- 4 Control jump height
- 5 Improve the feeling of a jump



TODAY'S GOAL

Turn our **physics system** into a playable **jumping system**.

Adding Gravity

In Session 6, we built:

ACCELERATION



VELOCITY



POSITION

Now we add gravity:

GRAVITY

`velocity.y``position.y`

Gravity is a constant downward acceleration.

```
velocity.y += gravity * dt
```

WHAT HAPPENS?

Gravity changes vertical velocity, which changes the player's position.



gravity

How Does a Jump Work?

Gravity changes velocity:

```
velocity.y += gravity * dt
```

To start a jump, we give the player an upward velocity:

```
velocity.y = -jump_strength
```

Remember:

Y increases → DOWN



Y decreases → UP

THE JUMP

JUMP

velocity.y = negative

MOVE UP

GRAVITY

velocity.y → 0

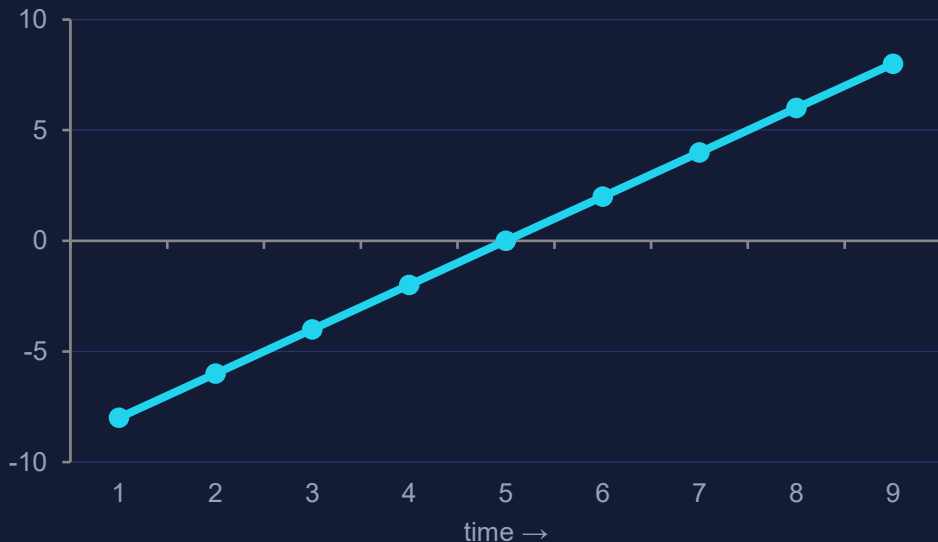
APEX

FALL

velocity.y → positive

Upward Velocity Becomes Downward

velocity.y during a jump (illustration)



`velocity.y < 0`

MOVE UP

`velocity.y = 0`

APEX

`velocity.y > 0`

FALL

CORE IDEA

Jumping starts with upward velocity. Gravity gradually turns it into downward velocity.

How Does the Player Actually Move?

We already know:

```
position += velocity * dt
```

For vertical movement:

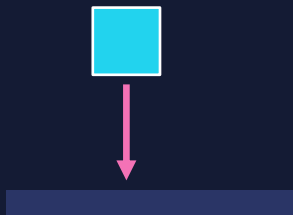
```
position.y += velocity.y * dt
```

When the player reaches the ground:

FALLING

↓ **GROUND**

velocity.y = 0



GRAVITY



```
velocity.y += gravity * dt
```



```
position.y += velocity.y * dt
```



PLAYER MOVES

Gravity controls the fall. Velocity controls the movement.
Position tells us where the player is.

Your Project

Start from your Session 6 result.

Turn the current physics movement into a playable jumping system.

1

Add Gravity

Make the player fall.

2

Add Jump

Press SPACE to jump.

3

Add Ground Detection

Make the player stop when reaching the ground.

4

Tune the Physics

Experiment with:

- `gravity`
- `jump_strength`
- `friction`



Test after every step.

BUILD



RUN



OBSERVE



MODIFY



REPEAT

Make It Feel Good

A working jump is not always a good jump. Experiment with:



LOW GRAVITY

- slower fall
- floaty feeling



HIGH GRAVITY

- faster fall
- heavier feeling

LOW JUMP STRENGTH



- small jump



HIGH JUMP STRENGTH

- high jump

Your Task: Find a combination that feels good to play.

Make Your Jump More Responsive

Once the basic jump works, choose one:

Variable Jump Height

SHORT INPUT → SMALL JUMP
LONG INPUT → HIGHER JUMP



short



long

Coyote Time

Allow the player to jump for a very short moment after leaving a platform.



PLATFORM



← grace period →



Challenge: Which mechanic makes your game feel more responsive?

Today's Big Ideas

GRAVITY



VELOCITY



POSITION

Remember:

Gravity changes velocity

```
velocity.y += gravity * dt
```

Velocity changes position

```
position.y += velocity.y * dt
```

Jump starts with upward velocity

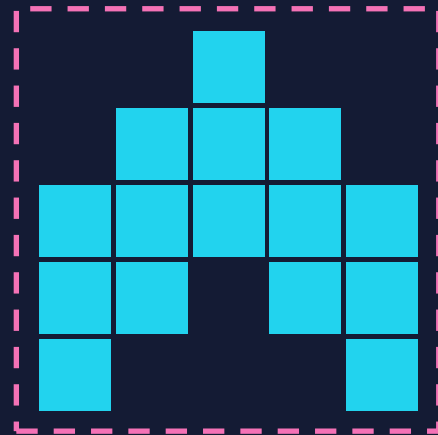
```
velocity.y = -jump_strength
```

The Goal: Don't just make the jump work. Make it feel good.

NEXT SESSION

Pixel-Perfect Collisions

Making collision detection more precise.



HITBOX

vs PIXELS